

Class 37 Least Squares Problems

Review: Write down in a sentence what you think is the primary problem people need to solve using linear algebra. What is this course all about?

The goal in this section is to find a vector $\mathbf{x}$ such that $A\mathbf{x}$ is as close to $\mathbf{b}$ as possible.

If $A\mathbf{x} = \mathbf{b}$ is consistent, then the $\mathbf{x}$ you want is just the solution to the equation. This $\mathbf{x}$ may or may not be unique.

If $A\mathbf{x} = \mathbf{b}$ is inconsistent, then the $\mathbf{x}$ you want is the solution to the equation $A\mathbf{x} = \hat{\mathbf{b}}$.

Since $\hat{\mathbf{b}}$ is in the column space of A we know the system $A\mathbf{x} = \hat{\mathbf{b}}$ will be consistent so there will be a solution.

We call this solution the _____.

Denote this solution by $\hat{\mathbf{x}}$.

We describe three methods to solve the system $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$ depending on the matrix A .

Method 1

Practice: Find a least-squares solution for $A\mathbf{x} = \mathbf{b}$ if $A = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix}$

Method 2: If the columns of A are linearly independent, we can use the QR factorization for A . In this case, the least-squares solution for $A\mathbf{x} = \mathbf{b}$ is given by

Method 3: If the columns of A are orthogonal, it is just easy enough to compute $\hat{\mathbf{b}}$. In this case, the least-squares solution for $A\mathbf{x} = \mathbf{b}$ is given by